MIL-C-20300
21 November 1951

MILITARY SPECIFICATION

CLOTH, CARTRIDGE, RAYON

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Army 50-11-108
5 March 1942

This specification consists of this cover sheet and Specification 50-11-108, dated 5 March 1942, attached hereto, without modification.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer .

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MIL-C-20300

U. S. ARMY
SPECIFICATION

No. 50-11-108
MARCH 5, 1942

CLOTH, CARTRIDGE, RAYON

A. APPLICABLE SPECIFICATIONS.

A-1. The following Federal and U. S. Army specifications, of the issues in effect on the date of invitation for bids, form a part of this specification:

No. 50-0-1, General Specifications for Ammunition Except Small Arms Ammunition.
CCC-T-191, Standard Specifications for Textile Materials.
No. 100-2, Standard Specifications for Marking Shipments.
DDD-S-751—Stitches; Seams; and Stitching.

B. GRADES.

B-1. This specification covers one grade of rayon cloth.

C. MATERIALS AND WORKMANSHIP.

C-1. *Material.*—Rayon cartridge cloth shall be woven from viscose rayon yarn containing no acetyl groups and not more than 0.2 percent of sulphur. The yarn may be approximately 275 denier and 120 filaments, having 3.5 turns per inch, or other viscose rayon yarn from which can be produced a cloth having the required characteristics.

C-2. *Workmanship.*—The finished cloth shall be uniformly and closely woven and shall be free from weighting materials. A plain single weave is satisfactory, but other weaves may be employed.

D. GENERAL REQUIREMENTS.

D-1. See section E.

E. DETAIL REQUIREMENTS.

E-1. *Appearance.*—Uniform and free from visible imperfections.
E-2. *Width.*—As specified by the contract. Minimum, 36 inches.
E-3. *Weight.*—Maximum, 12.5 ounces per square yard.
E-4. *Breaking strength.*—
Warp direction, minimum 300 pounds.
Fill direction, minimum 300 pounds.
E-5. *Slippage.*—When loads equal to 35 percent of the breaking strength in one direction and 50 percent of the breaking strength in the other direction are applied to seams in the cloth as directed under F-4d, the slippage of the yarn in either direction shall not exceed 0.25 inch.
E-6. *Stretch.*—In warp or fill direction, maximum, 12.5 percent.
E-7. *Ether extract.*—Maximum, 1.0 percent.
E-8. *Acidity or alkalinity.*—
Mineral, none.
Organic, maximum, 0.10 percent.
E-9. *Ash.*—Maximum, 0.2 percent.

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F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F–1. *Inspection.*—Inspection shall be in accordance with Specification No. 50–0–1 and shall be at the point of delivery unless otherwise specified in the contract.

F–2. *Size of lots.*—Maximum, 100,000 square yards.

F–3. *Sampling* shall be done as follows: Remove approximately 1 yard from each of 10 percent, but not more than 10 in number of the boxes comprising the lot. Label each of these primary samples so that the box from which it was taken can be identified. Make a composite sample of the lot, consisting of approximately ⅓ of each of the primary samples. Make all acceptance tests on the composite sample representative of the lot, using as many of the pieces as possible in making the tests. Place all samples in securely closed containers and label each container so as to show the name of the material, class, manufacturer, lot or purchase order number, and the number of yards in the lot. Hold the primary samples for possible future examination, should tests of the composite sample show that the lot does not comply with the requirements.

F–4. *Laboratory* examination shall be made as follows:

F–4a. *Width.*—Determine as directed by Specification CCC–T–191.

F–4b. *Weight.*—Determine as directed by Specification CCC–T–191.

F–4c. *Breaking strength.*—Determine by grab method as directed by Specification CCC–T–191 using material conditioned in accordance with that specification.

F–4d. *Slippage.*—Make this test of material which has been conditioned as directed by Specification CCC–T–191. Cut 10 test specimens as directed by Specification CCC–T–191 for use in determining the breaking strength, except that each specimen shall be 16 inches in length. Use 5 specimens for determining the slippage in one direction and the other 5 for determining the slippage in the other direction. For making this test, use a machine of the standard type specified by Specification CCC–T–191 for use in the determination of elongation. Adjust the distance between the jaws of the testing machine to 8 inches. Fold a test specimen back on itself so that the distance from the fold to one edge, parallel to the short dimension, measures approximately 5 inches and to the other, at least 10 inches. Take care to have the fold parallel to the crosswise yarns in the specimen. At 0.88 inch from the fold sew a seam parallel to the crosswise yarns of the specimen. Use a seam of the type designated as 301–SSa–1 by Federal Specification DDD–S–751, Stitches; Seams; and Stitching (seam SSa–1 as shown on page 27 of Specification DDD–S–751), formed with 1 row of stitch type 301, shown on page 5 of Specification DDD–S–751). Use 1 row of stitch type 301, 12 stitches per inch to make the seam. In making the seam use a suitable needle and a heavy silk thread of sufficient strength so that it will not break when the seam is tested for elongation as directed below. (A No. 21 needle and commercial grade H (14 lb.) waterproofed, Japan silk, fishing line thread with a left-hand machine twist has been found satisfactory). Place the test specimen, with the long 10-inch portion, uppermost, symmetrically in the upper jaws so that the long dimension of the 10-inch portion of the specimen is parallel to the direction of application of the load. Clamp the speci-

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men in the upper jaws so that the seam is at least 3 inches below the lower jaws. Attach a 6-ounce clamp (fig. 1) to the specimen at

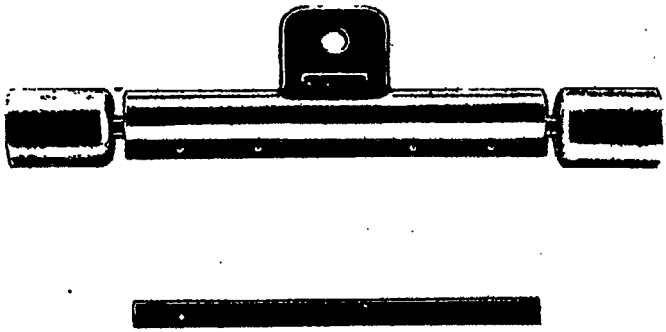


FIGURE 1.

a point beneath the lower jaws so that a uniform tension of that amount is applied to the specimen, and clamp the lower jaws. The clamp shown by figure 1 can be assembled from (*a*) a metal photographic film clip 3.75 inches wide, (*b*) a brass or steel rod 6 inches long and 9/16 inch in diameter, and (*c*) equal brass or steel weights which are sufficient to bring the total weight of the assembly to 6 ounces. Arrange the apparatus to obtain a tension-stretch curve up to the breaking strength of the specimen, the stretch being indicated in inches by the vertical component and the tension being indicated in pounds by the horizontal component. Start the machine and allow it to apply tension until the specimen breaks. Designate this graphic record as the fabric-stretch curve. Insert the same specimen in the jaws of the machine so that the seam is in a position parallel to the jaws and midway between the upper and lower jaws. Clamp the specimen in the jaws, applying a tension by means of the 6-ounce clamp as directed above before tightening the lower jaws. Arrange the apparatus so that the graphic record of this test will be made on the same chart and start at the same point. Start the machine and allow tension to be applied to the specimen until either the cloth breaks or the seam fails. Designate this graphic record as the seam-stretch curve. Calculate the slippage of the specimen as follows:

$$\text{Slippage, inches} = A - B - C + D$$

where

A=vertical components, in inches, of the seam-stretch curve at a tension equivalent to the required percentage of the breaking strength of this specimen indicated on the fabric-stretch curve.

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B=vertical component, in inches, of the fabric-stretch curve at a tension equivalent to the required percentage of the breaking strength of this specimen indicated on the fabric-stretch curve.

C=vertical component, in inches, of the seam-stretch curve at a tension of 1 pound.

D=vertical component, in inches, of the fabric-stretch curve at a tension of 1 pound.

Calculate the average slippage value in each direction.

F-4e. *Stretch test.*—Calculate the percentage of stretch for each specimen from the fabric-stretch curves, made as directed in paragraph F-4d.

$$\text{Percentage of stretch} = \frac{100\ (A-B)}{C}$$

where

A=vertical component, in inches, of the fabric-stretch curve at a tension of 50 pounds.

B=vertical component, in inches, of the fabric-stretch curve before a just perceptible load is assumed by the specimen.

C=length, in inches, of the portion of the specimen tested before a just perceptible load is assumed by the specimen.

F-4f. *Ether extract.*—Cut approximately 0.5 gram from each of the pieces comprising the sample and weigh all of the cuttings together. Transfer the weighed portion of the sample to a Soxhlet or other suitable extraction apparatus having a tared receiving flask, and extract the material with ether for 1.5 hours. Disconnect the receiving flask when most of the ether has collected in the middle section of the apparatus. Heat the receiving flask on a steam bath to evaporate any ether in the flask or accomplish this by means of a current of dry air. Heat the flask in an oven at 100° C. for 1 hour. Cool in a desiccator, and weigh. Calculate the increase in weight to percentage of ether extract.

F-4g. *Acidity.*—Cut approximately 0.5 gram from each of the pieces comprising the sample and weigh all of the cuttings together. Transfer the weighed portion of the sample to a 250 cc. beaker, add 150 cc. of neutral, freshly boiled distilled water, and boil this for 30 minutes. Keep the beaker covered with a watch glass during the boiling, and do not permit the volume of liquid at any time to be less than 75 cc. At the end of the 30-minute period add sufficient distilled water to bring the volume to 150 cc. and cool to room temperature. Add pieces of red and blue litmus paper which have been found to be sensitive to N/1000 sodium hydroxide and N/1000 sulphuric acid, respectively, when immersed in these for 3 minutes. At the end of 3 minutes note if the extract of the sample is neutral as indicated by the failure of either piece of litmus paper to be changed to the color produced by immersion of the paper in N/1000 acid or alkali. If the litmus paper is unchanged at the end of 3 minutes, filter the solution and titrate the filtrate with approximately N/10 alkali or acid, using phenolphthalein as the indicator. Calculate any acidity or alkalinity as acetic acid or sodium hydroxide.

F-4h. *Ash.*—Cut parts from each of the pieces comprising the sample so as to obtain a total of approximately 2 grams and weigh

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all of the cuttings together. Transfer the weighed portion of the sample to a tared platinum or porcelain crucible and moisten with concentrated nitric acid. Heat the crucible on a steam bath for 1 hour and then heat carefully over a flame so as to avoid any loss. When the contents of the crucible have been charred, heat to dull redness until all carbonaceous matter has been burned off. Cool the crucible in a desiccator and weigh. Calculate the gain in weight to percentage of ash in the sample.

G. PACKING AND MARKING.

G–1. *Packing.*—Rayon cartridge cloth shall be furnished in bolts wrapped in waxed paper and packed in boxes made of lumber not less than 5/8 inch in thickness. If preferred by the manufacturer, a box made of veneer will be permitted provided it has a border made from lumber not less than 5/8 inch thick and not less than 8 inches wide. No box shall contain more than 200 pounds of cloth.

G–2. *Marking.*—In accordance with U. S. Army Specification No. 100–2, with additional markings giving the net weight and lot number of the contents of each container.

H. NOTES.

H–1. *Uses.*—Rayon cartridge cloth covered by this specification is intended for use in the assembly of charges of propellent powder for cannon.

H–2. If the composite sample representative of the lot fails to pass the inspection tests, the manufacturer shall have the option of having tests of each of the primary samples made at his own expense. The manufacturer may then replace defective portions of the lot represented by the primary samples, which fail to meet the requirements and again submit the lot for inspection and acceptance, provided that the markings on the boxes are such that complete replacement of defective portions of the lot can be made to the satisfaction of the inspector.

H–3. The use of this specification, whenever applicable, is mandatory on all procuring agencies of the Army.

NOTICE.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may be in any way related thereto.

NOTE.—Copies of this specification may be obtained from the Office, Chief of Ordnance, Washington, D. C.

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